

BioRAG: Enhancing Biomedical Question Answering

Contrastive Fine-Tuning + Cross-Encoder Re-ranking

Group 2

BioASQ 12b Project Presentation

April 26, 2026

Biomedical QA

RAG

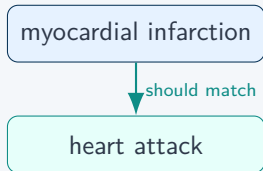
InfoNCE

Re-ranking

Motivation: Why Generic RAG Struggles in Biomedicine

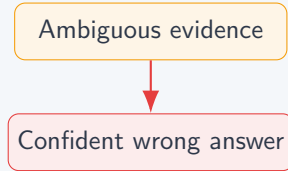
1. Dense retrieval gap

Generic embeddings often miss fine-grained biomedical equivalence or distinction.



2. Retrieval-induced hallucination

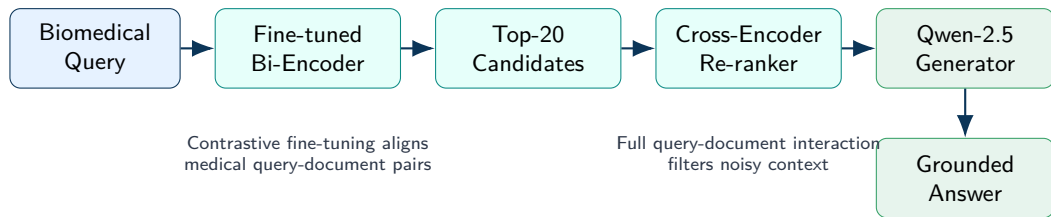
A marginally relevant document can push the generator toward a plausible but wrong answer.



Goal

Build a biomedical QA pipeline that improves recall while filtering unreliable evidence before generation.

BioRAG at a Glance



Core hypothesis

Embedding alignment via **InfoNCE** plus high-precision **re-ranking** improves biomedical QA fidelity.

Experimental Setup

Dataset

- BioASQ 12b biomedical QA task
- Stratified test sample: **500 queries**
- Question types: Yes/No, Factoid, List, Summary
- No overlap between fine-tuning pairs and test queries

Training details

- Retriever: PubMedBERT-base bi-encoder
- Epochs: **3**; learning rate: 2×10^{-5}
- Batch size: **16**; max length: **512**
- Re-rank candidates in batches of **32**

Official-style metrics

Yes/No	Accuracy, Macro-F1
Factoid	Strict Accuracy, MRR
List	Mean Recall, F-measure
Generation	ROUGE-L, BERTScore

500

stratified test queries

reproducible

Phase 1: Zero-Shot Baseline

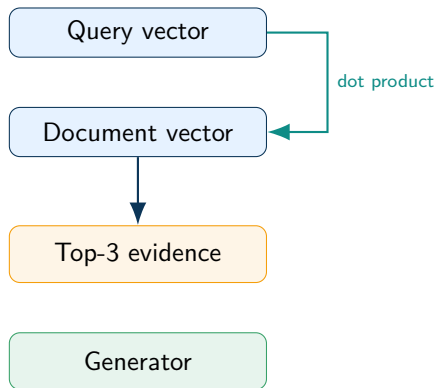
Frozen bi-encoder retrieval

$$s_{bi}(q, d) = E(q)^{\top} E(d)$$

- Encode query and document independently
- Retrieve top-3 from FAISS vector index
- Feed retrieved context directly into Qwen-2.5-7B-Instruct

Baseline weakness

Independent encoding is fast, but can miss subtle biomedical meaning and polarity.



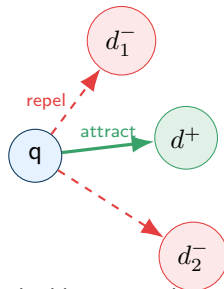
Phase 2: Contrastive Fine-Tuning

InfoNCE objective

For batch $B = \{(q_i, d_i^+)\}_{i=1}^N$, other positives in the batch become negatives:

$$\mathcal{L} = -\frac{1}{N} \sum_i \log \frac{e^{\text{sim}(q_i, d_i^+)/\tau}}{e^{\text{sim}(q_i, d_i^+)/\tau} + \sum_{j \neq i} e^{\text{sim}(q_i, d_j^+)/\tau}}$$

- Temperature $\tau = 0.05$
- Pull relevant biomedical pairs closer
- Push in-batch negatives apart



Embedding space becomes domain-aligned

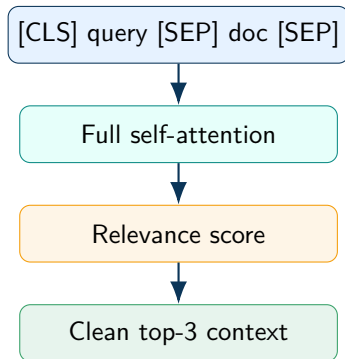
Phase 3: Cross-Encoder Re-ranking

Why re-rank?

Bi-encoders compress query and document into separate vectors. Cross-encoders score the pair jointly:

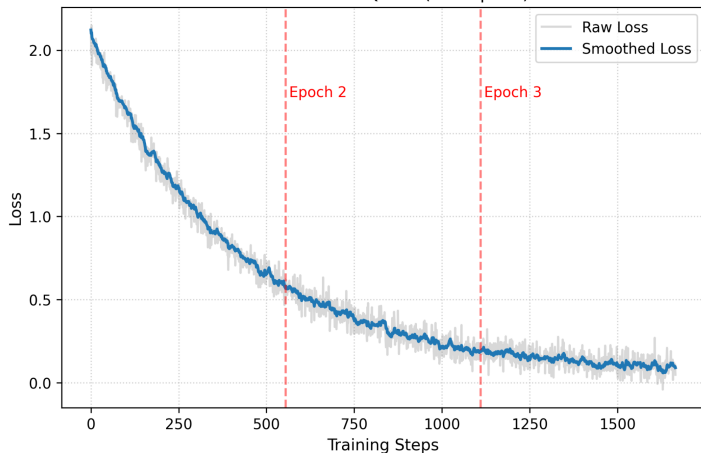
$$s_{cross}(q, d) = \sigma(W \cdot \text{BERT}([q; d]) + b)$$

- Retrieve **top-20** candidates
- Re-score query-document pairs with full self-attention
- Select best **top-3** evidence for generation



Training Convergence

Retriever Fine-tuning Loss (InfoNCE)
Dataset: BioASQ 12b (8887 pairs)



Interpretation

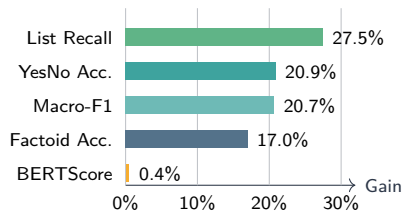
- Loss drops rapidly from about **2.2** to **0.5** in epoch 1
- Stabilizes near **0.1** by epoch 3
- No obvious overfitting pattern in the loss curve

Takeaway

The retriever learned a cleaner biomedical embedding space.

Quantitative Results: BioRAG Beats the Zero-Shot Baseline

Type	Metric	Baseline	Ours	Gain
Yes/No	Accuracy	0.6364	0.7692	+20.9%
Yes/No	Macro-F1	0.6137	0.7410	+20.7%
Factoid	Strict Acc.	0.3869	0.4526	+17.0%
List	Recall	0.3242	0.4133	+27.5%
Overall	BERTScore	0.8643	0.8679	+0.4%

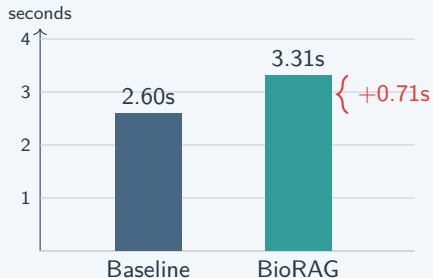


Best improvements

List Recall and Yes/No Accuracy benefit most from better retrieval and stronger evidence filtering.

Efficiency Trade-off

End-to-end latency



Why the overhead?

The cross-encoder performs full self-attention on 20 query-document pairs per question.

Why it is acceptable

Less than one additional second yields more than 20% gain on key accuracy/recall metrics.

Operational view

Good fit for asynchronous biomedical research assistance and evidence checking workflows.

Case Study: Safety Through Abstention

Query

Does physical activity influence gut hormones?

Truth: Yes

Zero-shot baseline

Answer: No

Retrieved a marginally relevant “bed rest” / inactivity document and generalized in the wrong direction.

Lexical overlap
without semantic polarity

BioRAG

Answer: I don't know

The re-ranker filtered weak evidence, so the generator abstained rather than fabricating.

High-precision evidence
as a safety gate

In clinical QA, abstention can be safer than confident misinformation.

Discussion: What Changed?

Retriever

Fine-tuning aligns medical concepts and improves coverage for entity-heavy questions.

Re-ranker

Cross-encoder catches subtle query-document interactions that bi-encoders miss.

Generator

Cleaner context leads to shorter, more direct, and semantically better answers.

Metric caveat

ROUGE-L may penalize concise factual answers, while BERTScore better reflects semantic improvement in this setup.

Limitations

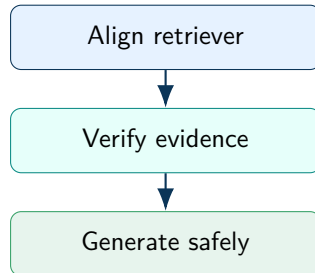
Cross-encoder latency, frozen generator, and evaluation on a 500-query subset leave room for future work.

Conclusion

Main claim

For biomedical QA, domain adaptation is not just an enhancement; it is necessary for reliable RAG.

- InfoNCE fine-tuning improves the retriever's biomedical embedding space
- Cross-encoder re-ranking improves evidence precision
- BioRAG achieves large gains in List Recall and Yes/No Accuracy with modest latency cost



Thank you! Questions?

References I



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